

Full marks are not necessarily awarded for a correct answer with no working. Answers must be supported by working and/or explanations. Where an answer is incorrect, some marks may be given for a correct method, provided this is shown by written working. You are therefore advised to show all working.

Section A

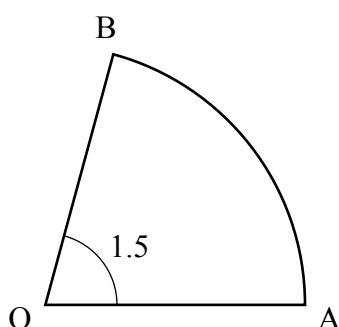
Answer **all** questions. Answers must be written within the answer boxes provided. Working may be continued below the lines, if necessary.

1. [Maximum mark: 5]

Points A and B lie on a circle with centre O and radius r cm, where $\angle AOB = 1.5$ radians.

This is shown on the following diagram.

diagram not to scale



The area of sector OAB is 48 cm^2 .

(a) Find the value of r . [3]

(b) Hence, find the perimeter of sector OAB . [2]

(This question continues on the following page)



(Question 1 continued)

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Turn over